

Ethical biotrade as a business alternative to biopiracy: three levels of interpretations of emerging social demands

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Summary:

An international convention now broadly ratified, the Convention on Biological Diversity identifies the trade of native ingredients and genetic resources as a tool to preserve biodiversity. This convention also defines the legal principles and criteria for this type of trade, involving the use of natural native ingredients, and often traditional knowledge hereof (articles 15 and 8j). Those sectors are also subject to a growing social demand for the respect of traditional knowledge and the rights of communities holding this knowledge.

This paper exemplifies and studies the contrasting frameworks that use to answer to those new demands, ranging from sustainability and corporate responsibility to fair trade, from fair trade to models contemplating the value and status of traditional knowledge. It also reviews the related certification systems and their *rationale*. Comparing those contrasting approaches and conceptual frameworks, it identifies the main lines of fracture and controversy, and links them to different approaches of intellectual property and related economic models.

Keywords: Convention on Biological Diversity, sustainable development, fair trade, ethical trade, evaluation.

Introduction

a. The legal setting

In 1992, the international community designed the Convention on Biological Diversity (CBD) as a new instrument to deter the loss of biodiversity worldwide. Signed by 193 countries, ratified by over 60, the Convention sets up three complementary objectives: the conservation of biodiversity, its sustainable use, the fair and equitable sharing of the benefit arising from their use. The latter is known as the “access and benefit sharing” mechanism, or ABS.

The development of sustainable supply chains tapping into the potential of biodiversity, especially from tropical countries, was thus viewed by the parts to the Convention as an instrument to enhance the protection of biodiversity. As the latter is more concentrated in the tropical fringe of our planet (counting for roughly 70 % of the species), negotiations at the Conference of Parties, gathered every two years within the framework of the Convention, have often been centred on mechanisms and good practices for “access and benefit sharing” (ABS), as the principles in the articles 15 and 8j of the Convention have been referred to.

The article 15 recognizes “the sovereign rights of States over their natural resources, the authority to determine access to genetic resources rests with the national governments and is subject to national legislation.” (§1). The following paragraphs are worth mentioning: “Access, where granted, shall be on mutually agreed terms and subject to the provisions of this Article.” (§4), “Access to genetic resources shall be subject to prior informed consent of the Contracting Party providing such resources, unless otherwise determined by that Party” (§5), “Each Contracting Party shall take legislative, administrative or policy measures, as appropriate, (...) with the aim of sharing in a fair and equitable way the results of research and development and the benefits arising from the commercial and other utilization of genetic resources with the Contracting Party providing such resources. Such sharing shall be upon mutually agreed terms.

Article 8 states: “Each Contracting Party [State] shall, as far as possible and as appropriate: (j) Subject to its national legislation, respect, preserve and maintain knowledge, innovations and practices of indigenous and local communities embodying traditional lifestyles relevant for the conservation and sustainable use of biological diversity and promote their wider application with the approval and involvement of the holders of such knowledge, innovations and practices and encourage the equitable sharing of the benefits arising from the utilization of such knowledge, innovations and practices”.

Further discussions reflect, with a North/South flavour, the diversity of views of countries and stakeholders (states, indigenous and local communities, companies, NGOs, etc.) on what is “prior informed consent” (right of access) and “fair and equitable benefit sharing”, two basic principles of the Convention. While the implementation of those principles by the States that are the parties to the Convention has been very slow and gradual, the United Nations Conference on Trade and Development (UNCTAD) opened up in the late 1990’s a Biotrade Initiative office in Geneva, which elaborated during the last decade a framework of principles and criteria for what has been later labelled as “ethical biotrade”.

b. Research questions and methodology

We used two complementary paths to gather information on the dissemination of the ABS principles of the CBD and their perception by private companies. For one part, we studied the emerging conceptual frameworks and regulatory schemes referring to those principals and to the issues they underline. For the other part, we observed the business practices of companies deeply involved in the relevant sectors. Both of those paths led us to a set of conceptual frameworks, used by the actors, and a set of related issues each underlined by one or several of those frameworks, and to the identification of the main lines of controversies between them. We then tried to understand what lies beneath those controversies, and found contrasting perceptions of intellectual property and business models.

The first phase of our investigation led us to assess the specificity of “ethical biotrade” as a conceptual framework emerging from a network of international institutions and private players, as compared to better-known frameworks, such as “organic agriculture” or “fair trade”. The United Nations Biotrade Initiative office has been instrumental in wording in 2006 a set of principles and criteria for ethical biotrade, after establishing a contact between the States and part of the private sector, mainly from the cosmetics industry. This conceptual framework was then slightly reworded and consolidated by the new Union for Ethical Biotrade (UEBT), a private sector initiative from the cosmetics and natural ingredient industry.

The second, and more substantial, phase was aimed at investigating the structure of operations of a variety of private companies of the biotrade sector. The methods we used were: literature review (scientific articles, company reports, press articles, web sites) and semi-directed interviews with private companies leaders of the cosmetic and food-supplement industries.

On a methodological level, our investigation is based on interviews of actors in Europe, South America and Southern Africa, reading of business reports, understanding of certification framework, and analysis of underlying issues. We started our investigation with companies met in several contexts. One of them was a conference organized by the Biotrade Initiative in April 2010, and subsequently companies participating in the Union for Ethical Biotrade. Others were small cosmetics or native food companies participating in the Fair Trade movement. Yet another was a cosmetics start-up company supporting the *Collectif pour une alternative à la Biopiraterie*, a French civil society group opposing biopiracy.

We then broadened our investigation basis to organizations in the source countries. South America and Southern Africa gave us a variety of cases to investigate. We thus discovered it was easier to access small and medium-sized companies than larger ones. Concerning large companies, Brazilian companies were more open to online interviews than French ones, which we thus plan to investigate later on.

1. From sustainable sourcing to ethical biotrade

1.1. Legal setting and issues related to the application of the CBD in Brazil

Brazil provides us with some of the most comprehensive cases of large cosmetics companies applying sustainability and ethical frameworks in their operations. The country is home to 15 to 20% of all fauna and flora species in the world, the Amazon being one of the main biomes of the South-American continent, together with the Andes. Access to genetic resource in the country is ruled since 2001 by the provisory decree 2.186-16/2001, amended by a series of ruling. The institution with the ability to give prospection and use permits is the Council for the Management of Genetic Heritage (local acronym: CGEN).

Implementation of the ABS protocol and the provisory decree in Brazil is renowned to be strict, with only a few dozen access permits for use of genetic resources and related traditional knowledge given in the past years (but hundreds for research and bioprospection). Other features of its application are worth to be mentioned however. Among them is the fact that the knowledge on many species from the Amazon, for instance, is not particular to one ethnic group, but has been widely shared and disseminated in the past centuries. This is why the application of article 8j of the CBD in Brazil doesn't preclude the use and trade of species that are widely known in the biome.

1.2. Beraca's and Natura's relationships with providers

The companies we have investigated were among those that are best established, and have implemented procedures to guarantee the sustainability of the resources used as well as an overall conformity to the Brazilian law, as part of their ethical *ethos*. Beraca is the main provider of natural ingredients to large Brazilian cosmetic companies O Boticário, Natura and Jequití. The main activity of Beraca, originally a chemistry company, is oil extraction, whereas Natura is the largest Brazilian cosmetic company, with an annual turnover of above 1 billion US\$. Natura also produces some of its own ingredients. Beraca's main Amazonian products come from fruits such as *acai*, *andiroba* and *buriti*, Natura's Ekos line uses *maracuja* (passion fruit), *mate verde*, *andiroba*, *cupuaçu*, *cocoa*, Brazil nut, *pitanga* and *buriti*.

Both companies source their ingredients through a network of *providers* organized in cooperatives, associations or informal groups, around 3 000 for Beraca, and 4 500 for Natura. During our investigation, in summer 2010, Beraca officials admitted that the relationships of the companies with its providers and local communities were not formalized yet, being under development. Its Biodiversity Enhancement program emphasizes sustainable use of biodiversity, as well as organic and Forest Stewardship Council certification (Beraca, 2009). It is thus one of the five Brazilian companies benefitting from a certification in organic cosmetics for 10% of its sales. Already 40% of its products is certified by Ecocert since 2006. Beraca is currently defining a policy for fair prices, and developing its relationship with communities, with the support of SEBRAE, an official institution supporting small and medium enterprises.

By contrast, Natura's relationships with its providers are highly structured. A Web portal defines the requirements and specifications for future or current providers of the company. All providers are regularly audited on quality, environment and social responsibility. A strict control from above is underlined. Natura's *2009 Sustainability Report* and several scientific articles allowed us to cross-reference certain claims.

The company clearly distinguishes different types of contractual obligations with its providing local communities, and the benefits that derive from them:

- *Agreements on access to genetic heritage or associated traditional knowledge*, which include benefit sharing under monetary and non monetary forma and a percentage of the product turnover. Natura has signed 26 such agreements, 25 in Brazil and one in Ecuador, each agreement involving a different community.
- *Local development projects*, which include investments from Natura in local communities to develop their capacities.
- *Partnerships with providers* that don't comprise ABS agreement, but support the production and harvesting of raw products, as well as the facilitation of relationships with transformers of raw products.

The following table, provided by Natura's 2009 report, gives us some useful information:

Table 1: Resources to the supplier communities (R\$ thousands)

	2007	2008	2009
Supply (sale of the raw materials)	863,6	2238,2	2756,1
Sharing of benefits from access to genetic resources or associated traditional knowledge	324,7	1136	1056,3
Funds and sponsorships	755,1	671,9	1137,7
Use of image	38,4	10,2	14,5
Training	49,9	18	151,8
Certification and stewardship	41,7	23,3	27,8
Studies and assistance	396,1	129,5	371,9
TOTAL	2469,6	4227,2	5516,1

Source : Natura, 2010, p. 51.

Several observations can be made: a) the last three lines are more related to the requirements of the development of the supply chain than to a redistribution policy targeting producers; b) the sales from the Ekos line of products has developed faster than the related payment for ABS; the latter being related to different products, which makes interpretation of this fact uneasy.

1.3. Biopiracy claims over Natura reflect complex local issues

Despite the ABS agreements implemented since 2006, and the company's claim to have implemented ABS-style agreements even before 2001, Natura has suffered from the strong involvement of certain parts of civil society and of the government for the defense of traditional knowledge on native resources. The controversies created by two of those cases cast an interesting light on sensitive and complex issues.

In 2001, women shopkeepers of the Belem Ver-o-Peso market were hired by Natura for a promotional campaign on local aromatic plants. Paid for the use of their image by the company, they later claimed a payment for giving access to their knowledge on those plants. Natura's defense relied on the alleged (and plausible) fact that knowledge on the use of those particular plants was common, or "diffuse, which is without a determined origin, and thus non regulated by the country's law" (Dias, A. and Weis, B., 2006). The implications of the concept of "diffuse knowledge", not a legal concept, if acknowledged officially, would have broader implications in the Amazon region, where, as mentioned before, the knowledge on many species of local plants are shared in the region, without this precluding the possibility of traditional knowledge on their uses, sustainable management or transformation.

In a later case, in 2007, concerning the *murumuru* plant and the traditional knowledge of the Ashininka Indians, two local cosmetics companies and Natura were trialed for having registered patents on the use and transformation of their product, while having benefitted from research on the matter done with the Ashinankas and a São Paulo-based NGO. The local companies were clearly responsible for this misconduct, but Natura was condemned with them to pay 50% of the gross benefits arising from the sales of those products.

In both those cases the controversy emerges not from facts, but from the interpretation of those facts. Here, specific actors classify the knowledge of a particular plant in different classes: common good for Natura in the Ver-o-Peso case, partially private and partially common (within research circles) for the *murumuru*. Private companies rarely acknowledge that this knowledge may be not a common but rather a collective good for specific people of the region.

Drawing a conclusion from those cases studied, we can say that the conceptual framework of Brazilian companies Beraca and Natura have evolved, in the last decade, from a strategy ensuring the sustainable and arguably ethical sourcing of ingredients¹ to access and benefit sharing agreements, as required by the implementation of the Brazilian law on access and use of genetic resources. This type of compliant sourcing and trade of local ingredients has been known recently as *ethical biotrade*.

2. Performance of voluntary ethical and sustainability frameworks

2.1. The emergence of principles and criteria for ethical biotrade

Both Beraca and Natura are part of a business alliance called Union for Ethical Biotrade (UEBT). The neologism *biotrade*, and later *ethical biotrade* originates in the activities of the Biotrade office of the United Nations, based in Geneva. The Biotrade Initiative is in fact a shared initiative from the secretariat of the CBD and the United Nations Conference on Trade and Development (UNCTAD). Its aim is to "contribute to the conservation and sustainable use of biodiversity through the promotion of trade and investment in BioTrade products and services in line with the objectives and principles of the Convention on Biological Diversity (CBD)" (BioTrade Initiative, 2007). Its perspective is to conciliate the interests of private companies with the rights of rural and indigenous communities under the CBD, as well as to mobilize private and institutional actors to develop international markets for genetic resources.

¹ For instance, by implementing local resource management plans, ensuring a constant price and social benefits to providers, as well as pursuing organic and FSC certification.

In 2006-2007, the BioTrade Initiative defined the *BioTrade Principles and Criteria*, a framework for relationships between private companies and local communities comprising 7 principles. This framework was revised and remodelled by the new Union for Ethical Biotrade, created in May 2007 with the support of UNCTAD and the International Finance Corporation. The result is the “BioTrade Verification Framework for Native Natural Ingredients”, which contains the following definition:

The term “native” applies only to species collected or produced in their original distribution areas. Native natural ingredients may include; for example essential oils: *Lippia alba* (cultivated), *Bursera graveolens* (wild), vegetable oils, such as: Shea butter (wild in community lands), *Oenocarpus* oil (wild) and colorants, like *Genipa americana* (wild). The Verification Framework does not address naturalized species, that is species introduced via human means outside their natural area of distribution.

2.2. Principles of ethical biotrade

The UEBT framework comprises the following principals, derived from the CBD, as well as other international conventions and normative references: Conservation of biodiversity, Sustainable use of biodiversity, Fair and equitable sharing of benefits derived from the use of biodiversity, Socio-economic sustainability (productive, financial and market management), Compliance with national and international regulations, Respect for the rights of actors involved in BioTrade activities, Clarity about land tenure, use and access to natural resources (*UEBT, 2007*).

We have noticed some minor limitations to this framework. For instance the list of conventions referred to is limitative (no reference to ILO convention 169 on the rights of indigenous people), the order and importance of principles criteria emphasize the socio-economic and environmental performance before the respect for the rights of actors and the informed consent of local communities. However, it still offers the first major voluntary comprehensive framework referring to the CBD.

2.3. Coverage of legal and sustainability issues by voluntary frameworks

Our research contemplates a comparison of voluntary ethical or sustainability frameworks (most are known as “labels”) used in the industries using natural ingredients, regarding their coverage of issues related to sustainability, and to the compliance with the CBD. We set up a typology of voluntary frameworks (labels and guarantee systems) in 5 groups: organic agriculture, organic and ecological cosmetics, fair trade, sustainable forest products, and biotrade.

We then identified the main issues covered by those frameworks, which led us to identify two fundamental axis included in the concept of sustainable development. For each axis, we identified as well a basic level of understanding of the issue, and an advanced level.

The first axis is related to the so-called “natural environment”, which comprises a basic level commonly worded as “the respect of environment”, and an advanced level defined as the integration of biodiversity in the companies’ operations. The second axis is related to socio-economic issues. Its basic level is to cover sustainable conditions of production; its advanced level is defined by conditions of fairness and equity between partners of the supply chain. The respect of the principles of the CBD makes a third axis, only contemplated by a few frameworks.

In the current state of our research, it was not possible to assess the *performance* of the voluntary frameworks contemplated, but only the way they ensure or not a coverage of those issues in their sets of principles and criteria. An assessment of the performance of each framework, as regarding the contemplated issues, would require significant further investigation. The following table summarizes the results of our investigation on coverage:

Table 2: Coverage of legal and sustainability issues by voluntary frameworks²

	Environment	Biodiversity	Conditions of production	Equity in trade	PIC and ABS
Organic Agriculture					
Organic label	●				
Private more restrictive OA labels	●	○			
Cosmetics					
Cosmos	●				
NaTrue	●				
Fairtrade					
Fairtrade	●		●	●	
ESR (Ecocert)	●		●	●	
FTO Mark (WFTO)	●		●	●	
SGAP Minga	●		●	●	
Forest products					
FSC	●		●		○
Forest Garden Products	●	●			
Rainforest Alliance	●		●		
Biotrade and new labels					
UEBT audits	●	●	●		●
FairWild Label	●	●	●	●	●

One observation is that the most renowned frameworks, such as *organic agriculture* and *fairtrade* labels, as well as labels related to cosmetics, don't take into account the principles of the Convention on Biological Diversity. Rainforest Alliance covers several issues, but is usually perceived as weak in its applications. Organic agriculture clearly covers neither socio-economic conditions nor CBD principles. Forest Garden Products is a new label created by the International Analog Forestry Network. Its environmental requirements are quite high, including many biodiversity aspects, but it doesn't include socio-economic or CBD principles.

The Forest Stewardship framework, first set up 15 years ago, contemplates *non-timber forest products* as well as wood products. Though it does not refer precisely to the terms of the CBD, it is the only framework of its generation to comprise explicit reference to the rights of indigenous

² A full dot means that the issue is clearly integrated into the framework. An empty dot means it is only partially integrated.

communities, including prior informed consent and fair “compensation” for the use of their traditional knowledge on tree species and traditional management systems.

Our coverage table thus shows quite a few empty cells for most of the frameworks studied. Despite its cost, certification is not always relevant to the whole set of issues consumers may want to have a guarantee for. This situation often obliges producers to pay and get two labels instead of one, at double the price (for instance: Fairtrade and organic). Some schemes have been set up to build more global certification (Bioequitable in France, among others), without meeting a large success. The reasons may be twofold: different segments of consumers emphasize some set of issues above others (organic above fairtrade or fairtrade above organic, for example), and certification agencies’ interest is to maintain a proper segmentation of their offer.

2.4. Constituency of UEBT and Fairwild

In this landscape, the UEBT and FairWild (accreditation programme for wild collected products) frameworks, latecomers, seem to cover quite a large range of issues. They are currently the only two frameworks explicitly integrating the legal constraints of the CBD, which in itself cuts across issues related to the biodiversity, socio-economic and legal condition of trade, and resource management. Both of those frameworks thus anticipate on an emergent demand of biotrade for native products that respect legal as well as ethical and sustainability principles.

It is time to present these new actors in the landscape of ethical and sustainability frameworks.

The Union for Ethical Biotrade defines itself as an organization of members. Its bylaws identify three types of members: trading members, affiliate members and provisional members. It counts currently 20 trading members, among which: Aldivia (France), Aroma Forest (Madagascar), Beraca and Natura (Brazil), and Swazi Indigenous Products (Swaziland), all mentioned at some point in this article. Listed affiliate members (non trading) are 15, among which are for instance: the International Finance Corporation, the International Union for the Conservation of Nature, Fondo Biocomercio (Columbia) and PhytoTrade Africa, which we will mention further in this paper. Five Provisional members have successfully passed the UEBT entry audit and are in the process of developing their work-plan to reach compliance with the Ethical BioTrade Standard.

Most UEBT board members are representatives of the private sector in Europe, Africa or Latin America. Civil society is represented mainly by two affiliate members: IUCN and NGO L’Homme et l’Environnement, active in Madagascar.

All members must comply with minimum indicators of the UEBT verification framework, and pay an annual fee based on their turnover or budget, ranging from 0,05% to 0,1% of the turnover for trading members, and close figures for NGOs, with maximum thresholds. Though its evaluation framework and its governance is transparent, it is clear that UEBT, which rules the first ethical biotrade framework, remains an organization meant for the private sector.

FairWild, by contrast, operates as other voluntary certification bodies, being open to any type of organizations, including research institutes and private companies. It aims to fill a gap in existing certification, by providing principles and criteria adapted to the collection of wild-collected natural ingredients and products thereof. Its framework is inherited from the standard

elaborated by a FAO workgroup in the 1990's and 2000's, the International Standard for Sustainable Wild Collection of Medicinal and Aromatic Plants (ISSC-MAP).

In 2009 FairWild took the ISSC-MAP principles, and organized them in a comprehensive framework, updated in 2010 (*FairWild, 2006 and 2010*). These standards facilitate the implementation of the 1993 recommendations of WHO/IUCN/WWF and the WHO 2003 good practice guidelines for the conservation of medicinal plants. FairWild claims its affiliation with fair trade, and its principles and the related criteria are close to those of Fairtrade International. The 2009 standards however required a stricter management of price premium than the Fairtrade standard itself.

While FairWild defines itself as an independent body, its position is that of a niche certification, dealing with the sourcing of wild ingredients only. By contrast, the UEBT is member-based, and its framework deals with the whole supply chain.

3. Innovative business models

Far from the spotlights of United Nations offices and the marketing force of major cosmetics or food companies, several small or medium companies have developed innovative partnerships with a few local indigenous communities. These businesses are more interested in developing positive relationships with their partners than in showing off labels.

An in-depth study of three of those companies, two of them sourcing products from South America, and one established in Southern Africa, shows how the respect of local communities and of their traditional knowledge can be embodied both within fair trade principles and within new ethical business models.

3.1. Updating the fair trade software: the warana partnership

The partnership developed between the Amazonian people Satéré Mawé and two European fair trade companies exemplifies in many ways one of the most ethical and sustainable trade relationship between an indigenous people and private companies that we could observe for this study.

The Satéré Mawé have been for at least three centuries the custodians of the *guarana* (*Paulinia cupana*), a plant which is at the origin an emblematic drink in Brazil. The fruit of this climbing plant originating from the Amazon forest is harvested, its seeds roasted and consumed by the Satéré Mawé people. This community, which comprises today around ten thousand individuals, calls itself "Sons of the Warana", the latter being the original indigenous pronunciation for *guarana*. The powder issued from the roasted seed is of a highly symbolic importance. It is consumed within the family and at community gatherings.

The Satéré Mawé have preserved and adapted with ecological methods this plant within their territory, which is now a reservation of 780 000 hectares. The production is non-intensive, pest control is done by biological techniques, and fertilization by a local stingless species of bee. However most Brazilians have never heard of the Satéré Mawé. The largest part of the production is developed outside of its ecosystem, even outside the Amazon, in the state of Bahia, by transnational drink corporations. The national service for small and medium enterprises, SEBRAE, thus supports the so-called conventional production of guarana, which includes a technological pack with hybrid seedlings, chemical fertilizers; it also tests transgenic seedlings.

The Satéré Mawé have developed a unique know-how for all steps of the production³, making the *Warana dos Satéré Mawé* a unique and distinctive product, for which it is currently seeking recognition through a geographical indication. Community cohesion, strategic vision of its leaders and partnerships with European fair trade companies are the main factors that have favoured over the past 16 to 17 years the revival of this tradition and its role as the keystone of the Satéré Mawé's endogenous development project.

The trade relationships between the two main partners of the Satéré Mawé producers, French specialized company Guayapi and Italian fair trade purchasing company CTM have contributed to support the autonomy of the local producers and its community. In both cases, the fair trade philosophy was a good foretaste of the balanced trade relationship maintained on the long-term with the Satéré Mawé.

Fair trade principles include the following: long-term and transparent relations, fair price, prefinancing, democratic organizations and capacity building. Both European partners followed its usual practice for special products to set up the price at the level of the producers' offer. Bought at prices 4 to 7 times higher than other qualities of guarana on the Brazilian market, Guayapi CEO and founder Claudie Ravel, insists that the Satéré Mawé warana and the conventional guarana are totally different products, due to the questionable quality of conventional guarana, and the distinctive cultivation and transformation methods of the Satéré Mawé. Those methods are now casted in a guide in Portuguese and Satéré Mawé language.

Artisans du Monde, which sells the warana-based soda of CTM, and Guayapi both publicize the breakdown of the buying price for warana. A detailed table and analysis is available in our study (Johnson, P., 2011). We will just mention here the importance of two lines in this breakdown: the part going to the warana producers (between 26% and 35%), and the part going to the community for concepts of local and community development (23,4 to 29%). The rest of the price goes to technical support (around 10%), transportation and taxes (9%).

In the past years, the part going to the community has been managed in different ways, due to the development of capacities of the producer group, the Satéré Mawé Producers Consortium. Initially, it was directly perceived by the political body, the General Council of Satéré Mawé Tribes. The principle of an important contribution of the sales of warana to the development project of the Satéré Mawé has been maintained through those changes.

For the purpose of this article, is it is useful to mention that while both companies have of course the legal permits to import warana, their interpretation of this financial contribution to community development don't refer to the CBD and its concepts of "prior informed consent" or "access and benefit sharing", but to fair trade concepts exclusively. For David Erhart, who did an extensive study on the contrasting guarana supply chains for Artisans du Monde (*Barthel T. et Erhardt D., 2009*), the portion of the price going to local and community development is a fair trade premium, a concept applied to all products fair traded products, be it a commodity, handicraft, or (as in this case) a higher value-added product. In his view, followed by other fair trade organizations, it is not a specific contribution to the use of traditional knowledge.

³ from the culture and extraction of the seed, to its smoking with specific aromatic wood species during three months on large clay pans, followed by its breaking in powder with wood pestles.

The *warana* cases highlight the fact that companies' communications refer to specific universes. The cultural universe of CTM, Guayapi, Artisans du Monde and their customers being that of *fair trade* rather than *corporate responsibility*, referring to CBD and its principles seemed less relevant to them than referring to more familiar concepts. Like Molière's M. Jourdain, Guayapi, CTM and Artisans du Monde may have been doing PIC and ABS without even knowing it.

However, both have been proactive in the valorisation of the traditional warana methods in European markets, and in broking partnerships with other organizations. A result was that Satéré Mawé's Warana has been incorporated in Slow Food's Ark of Taste catalogue, which refers to "small-scale quality productions threatened by industrial agriculture, environmental degradation and homogenization". The product also gained Forest Garden Products certification, an emerging label mentioned previously in this article.

3.2. Developing a new ethical model with Shamans' knowledge for the luxury market

In the following case of trade in cosmetic ingredients, an ethical business model compatible with the respect of traditional knowledge and the engagement against biopiracy was formalized by the company leader himself. This model highlights the research and development done by indigenous communities themselves over the centuries. Exchange and inspiration between the company, indigenous communities and also the government of Peru was on both sides.

Savoirs des Peuples⁴ is a French ethical start-up of the cosmetic sector, created by Daniel Joutard. After having worked for several years with indigenous organizations in the Andean part of South America, D. Joutard saw the development of a luxury cosmetic product range as a means to valorise the knowledge and the work of those communities in Peru and Ecuador. "There is no more need to demonstrate the knowledge indigenous people have of plants. (...) Some Yaneshas and Shipibos Shamans as well as healers from Peru know several hundreds of plants and their use." D. Joutard further highlights the value of this knowledge for private companies, which can thus accelerate their research and development processes (Joutard, D., 2008).

The species involved in the Aïny range, the current brand of Savoirs des Peuples, are the following: *Sacha Inchi* from the Ashaninkas in Peru, *Ungurahua*, harvested by the Shuars in Ecuador, *Molle* and *Achiote* of the Andean region. *Sacha Inchi* is rich in omega 3 and 6, *Ungurahua* in omega 9. The company's current motto is "the Powers of Sacred Plants", and it refers explicitly to the local Shamans health and cosmetic knowledge.

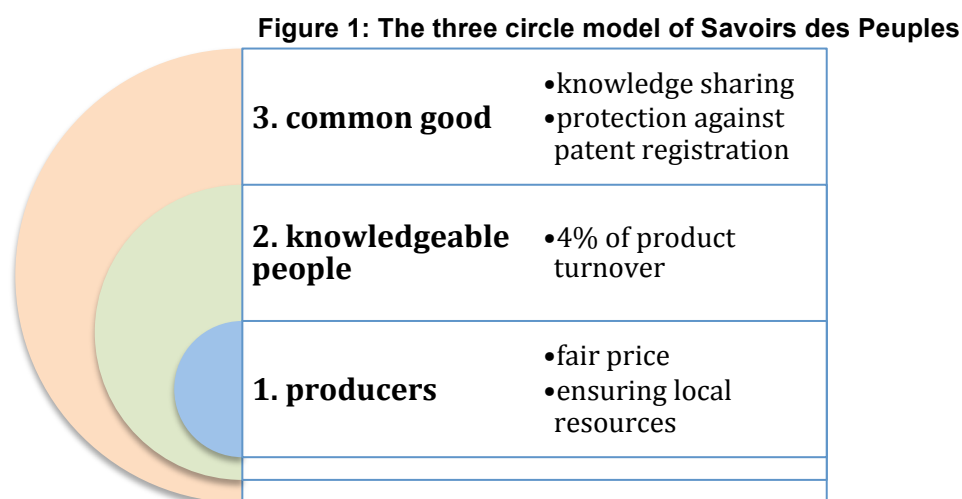
Savoirs des Peuples defends a strict interpretation of prior informed consent for the commercial use of any traditional knowledge related to plants, as well as a shared interpretation of benefit sharing. "Biopiracy starts when this sharing becomes theft, that is when the study of traditional knowledge is done hiding the commercial objective, or when selling is done without the consent of traditional people and a sharing of benefits accepted by all parties." (Joutard, D., 2008). For him "any use or company patent is questionable, even if it is not directly linked to the functional usage of the plant, because the company's product is a construction, in which the indigenous communities play a major part."

Having received prior consent, the company proposed the following conditions to the indigenous community as a whole: the payment of 4% of the global turnover of the product for

⁴ Savoirs des Peuples means « People's knowledge » in French.

the concept of use of its traditional knowledge, the engagement not to register any patents on the plants used or any of its extract or transformation, and the publication of the company's findings in a scientific journal, as a means to impede any patent registration on those products by other companies (Rousseau, C., 2009).

Thus, the model developed by Savoirs des Peuples builds on the principles of the CBD, and goes beyond a fair trade price approach, or a compliance and certification approach. It has been formalized by its author as a “three circles model”, of which we suggest the following representation:



Together with the French Workgroup for an alternative to Biopiracy (Collectif pour une alternative à la Biopiraterie) D. Joutard defends an approach rejecting the use of patents. As we shall see hereafter, this is not however the case of all companies involved in ethical biotrade.

4. Emerging areas of controversy

Southern Africa gave us an interesting field for comparing several contrasting approaches to *ethical biotrade*. This region of the world is very rich in plant and animal species, as well as indigenous and traditional people, sedentary or nomad. The diversity of plants is estimated at around 30 000 species, of which only 50 are currently traded. Many locally produced and sometimes traded species are unknown outside Africa, having for some of them high commercial potential.

Several biopiracy cases have been pinpointed by civil society in the past years, such as the cases of *Pelargonium sidoides* or the genus *Hoodia* from Namibia. The South African NGO African Centre for Biosafety monitors such cases, informing their partners in other countries when specific cases arise.

4.1. Two contrasting approaches of ethical biotrade

Marula (*Sclerocarya Birrea*), by contrast, is a plant common to the region, from which women extract a cosmetic oil, sold locally and for exportation. In the past ten years, two strikingly different approaches have been developed by successful local organizations to market this oil. The contrast reveals strategic issues, such as: the relevance or irrelevance of patent registration

for ethical biotrade, and the capacity of local producers and companies to transform the product and reach export quality.

Phytotrade Africa (PhA), the *Southern Africa Natural Products Trade Association* operates as a facilitator for the production and trade of “products derived from plants indigenous to Southern Africa; they include foods, drinks, oils and ingredients used by the food and cosmetics industries.” Registered in 2002 as a non-profit organization, it is membership-based and represents private sector businesses, development agencies, individuals and other interested parties, in eight Southern African countries (PhytoTrade Africa, 2007).

In the case of marula, PhA developed a partnership with French ingredient company Aldivia and the Eudafano Women’s Cooperative, comprising about 4 800 women in Namibia, for the production and marketing of marula oil. This action led to the registration of a joint patent by Aldivia and PhA on an extract of marula, under the commercial name Maruline®. This patent registration was done after many internal debates and tensions between the four partners (local cooperative, PhA, Aldivia and a support NGO)⁵.

PhA’s developed its position on benefit sharing, the issue of patents, and bioprospection, in two documents made public by the association (PhytoTrade Africa, 2003 and 2007). “In early 2003 it published a set of Bio-Prospecting Guidelines for its members, the aim of which was to advise African producers how to respond to bio-prospecting approaches from other companies.” Its interpretation of bioprospection is narrow, as evidenced in Annex A “Best practices in benefit-sharing - by type of industry”. For “Botanical medicines, food supplements, personal care and cosmetic products”: “genetic resource access is often de-linked from benefit-sharing in that the genetic resource may be treated as a commodity and/or traditional knowledge is acquired from literature and databases.”

In its position paper on ABS, PhA thus states: “However in practice, there is little traditional knowledge that is patentable, since much of it has emerged in the public domain through the work of ethno-botanists and other researchers over the last century or so.” In the areas where traditional knowledge may lead to patentable applications, “it is of course important to try and maximise the benefit- sharing opportunities that could arise from such patents.” Indeed “PhytoTrade Africa recognises that patenting is an important commercial strategy”.

The problem is this position creates strongly limitations on the right of indigenous people to prior informed consent, and puts them in a position of weakness while negotiating benefit-sharing arrangements. It is similar to Natura’s difficult position about “diffuse knowledge”. PhA is mainly interested in the development of local companies of the natural ingredients and products sector, whose interests are not necessarily aligned with that of local communities. Its position that “Benefit-sharing as a concept is irrelevant if there are no benefits to share”, is not defensible in front of the CDB and its transcription in national laws.

In the case of the Eudafano Women’s Cooperative, the patent on Maruline® represents a danger for this cooperative of being tied on the long term to the company and organization registering the patent, and making any diversification of its clients difficult. Their joint 2005 press release announcing “Maruline®, the first African active botanical ingredient from fair trade and

⁵ Information from a French expert who was working in Namibia on the marula supply chain during this exchange between partners.

sustainable sources” is thus a clear distortion of Fair Trade principles, which advocate for a diversification of producers’ clients.

Proof that other strategies are possible within the ethical biotrade movement, the young non-profit Swazi Indigenous Products (SIP) company, instead of speculating on the benefits of patents for local communities, followed this other advice of PhA: “increasing the local value also provides a basis for source-country groups can (sic) develop local markets for processed products, which are generally less unstable than international botanical medicine markets”. The main distinctive feature in the constituency of SIP is that it is owned by local producers groups, comprising 2 600 producers of the Belt Land region in Swaziland.

A member of UEBT, PhytoTrade Africa, the World Fair Trade Organization and its African chapter COFTA, SIP could contemplate different business strategies within those networks. Its own strategy is to position itself both downstream and upstream of the supply chain, and to market its products under its own brand. Local markets still represent 75% of the production, and marketing in Europe and North America is done by partner companies.

The testimonies given by women and audited by Fairtrade International put in evidence the additional income and the power that women could generate through the collection and marketing activity of the company. Premiums of 4% are paid to women that are member of the company (for a fee of about 1 €), and 8% to those collecting in wild pasture areas a few kilometres from the villages, as this allows the product to be certified organic. No patent has been registered by SIP on any of the products it works with (*marula*, *ximenia*). For its director, John Pearce: “It is the story behind the product that make its value”.

4.2. Lessons learned from the marula cases

The Southern Africa cases are illustrative of a touchy issue within the ethical biotrade movement. Before we summarize this issue, it is worth noticing that both organizations are members of UEBT and have initiated their operations with funds and research from development cooperation.

While Phytotrade Africa focused on development through a business-as-usual strategy contemplating patents and marketing of often classical research and development efforts marketed as private innovations, Swazi followed a bottom-up approach reinforcing local capacity and allowing local stakeholders to capture added value. The Director of SIP sees patent as non-necessary, especially when it involves companies from importing countries. With a different history and other premises, this organization’s position on this issue is shared with that of Savoirs des Peuples and its three circles model.

There currently is a lack of empirical evidence of cases where patents would be a positive or a necessary step for the respect of local indigenous knowledge and the equitable sharing of benefits. Quite on the opposite, it can be argued that patents value a few months or years of technological research and development efforts by private companies above the century-long, powerful and quiet, building-up of indigenous knowledge on local ecosystems. Anthropological research shows that a continued presence of the same people over 5,000 years in a specific area may be necessary for it to create a corpus of practical and theoretical knowledge on the local ecosystems. That is a far longer time scale than the product development-cycle in any contemporary economic sector.

5. Conclusions

5.1. Contrasting frameworks for biotrade

The panel of cases presented above shows the co-existence of at least three ethics and sustainability conceptual frameworks for the relationships between local communities and companies trading native ingredients. We can summarize the frameworks identified in the course of our investigation as follows:

Sustainability and Corporate Social Responsibility framework

Private companies such as Brazilian Beraca or Natura make a privileged use of the language of Sustainability and Corporate Social Responsibility to describe the organizing principles for their supply chains and their ethical relationships with ingredients suppliers.

Their main focus is on sustainable management of their supply chain, and on compliance, which involve: resource management, social benefits, signing ABS agreements. In this model, intellectual property usually remains the ownership of private companies.

Union for Ethical Biotrade principles and criteria

This recently designed framework rigorously gathers the principles (mentioned above) and criteria to which companies must comply, when dealing with native natural ingredients. It contemplates the principles of the CBD, among which prior informed consent and benefit sharing. “The use of traditional knowledge shall be recognized, promoted and adequately compensated” (§3.4.). No specific regime of intellectual property is promoted.

Fair Trade principles, as applied to native ingredients

The Fair Trade movement is mainly concerned by the development of communities from the South through balanced trade relations. It under-estimates or fails to identify many contemporary issues, among which the importance of intellectual property rights, the value of traditional knowledge, and other aspects of the CBD. Some specific fair trade companies have a more relevant insight of those issues, as reflected by their particular partners. However, they tend to translate those issues into the fair trade conceptual framework and vocabulary exclusively (“development premium”, among others).

Daniel Joutard’s three circles model

This innovative model, with practical implications on the relationship between trade partners, involves three concentric actions for the company dealing with natural ingredients and traditional knowledge of local communities, viewed as partners: Fair price to producers, 4% of the product turnover for the use, with its consent, of traditional knowledge to the representatives of the community, and publication of the results of R&D on those products, to prevent the registration of patents. Traditional knowledge is viewed neither as a good that can be privatized, even through benefit sharing, nor as a common good of humanity, but as a collective good for its people.

Table 3: Summary of the frameworks contemplated in this article

	Examples	Conceptual framework	Main focus and actions	Perspective Ethos	Intellectual property rights
Sustainability Corporate Social Responsibility	Beraca Natura	Sustainability Ethics Compliance	Resources management Benefit sharing	Sustainability Ethics	Patent model is dominant
Union for Ethical Biobased principles and criteria	PhytoTrade Africa	Sustainability CSR, ethics Compliance, including CDB provisions	UEBT framework Conservation, Labour and Community Rights Benefit sharing		Members are divided on use of patents
Fair Trade principles	Fair Trade companies WFTO	North-South divide Communities from the South are producers of commodities. Fair Trade premium	Transparency Fair price Fairtrade premium	Partnership Equity	IPR not considered in model and standards
Fair trade relationships, in specific cases of native ingredients	Sateré Mawé's European partners: Guayapi CTM AdM	Fair Trade supports partners from the South's development project.	Fair Trade principles. Branding the product's value added through specific communication and adequate labels		Traditional knowledge respected and valued.
Three circle model	Aïny Could apply to the above, to Swazi IP	Value and knowledge comes mainly from local communities.	• Fair price • 4% turnover for TK use • publication of R&D -no patents	Partnership Equity Value creation	Patents rejected by model. Collective goods considered

5.2. From Ethics and Sustainability to Fair value creation

The models identified in this article are not opposed in their principles one to another. Rather, they focus on different concepts, thus contrasting in their perspectives, or *ethos*, which range from attention to Sustainability and Ethics (Beraca, Natura, and other major actors of the cosmetics industry), to the building of partnerships based on fair trade relations (ordinary fair trade relations), and finally the valuation of traditional knowledge as a collective good.

In an ideal world, those models would be seen as different levels of elaboration of ethical and sustainable principles in the biobased sector. In practice, the difference of *ethos* means some areas of controversy remain. As developed above, an issue that crystallizes opposing positions is that of the ruling of intellectual property rights related to plants and traditional knowledge.

This issue is not mainly legal. The CBD does not clearly arbitrate on it, nor does the World Intellectual Property Organization yet. Not surprisingly, different countries have different positions regarding the use of patenting in biobased. Those with Biodiversity Laws tend to monitor patents registration on their native species in main Patent Offices with close attention.

This is the case of Brazil, Peru and other countries of the Andean Pact, of India, China and the Republic of South Africa.

Patent registration is clearly a tool to secure intellectual property rights that is more adapted to the operations of private companies than to traditional knowledge and their custodians. Countries like India, China and Peru have begun to register traditional knowledge in special registers or databases, with a variety of options left after registration. Most major companies registering patents do so to secure their business from competitors, more than to share value with their partners. PhytoTrade Africa and Aldivia pretend to do the same in the Maruline® case.

For companies involved in closer partnerships with indigenous people (Guayapi, Savoirs des Peuples) patent registration on native products tend to be synonym of “biopiracy”, as most of these people have a very different representation of intellectual property. They are generally opposed to patents related to natural resources and traditional knowledge, even when schemes of benefit-sharing or co-ownership of those patents are proposed to them.

The issue even cuts across UEBT membership, some members resorting to patents, others apply other means of “value creation”, such as increasing the value locally, by preserving and valuating traditional knowledge, and local processing of native ingredients. Companies following those business models create value with their local partners without needing to register patents.

5.3. Perspectives on social acceptability of patents and emerging models

From a political perspective, the patent controversy reflects the difficulty of building consensus on how the principles of the CDB should be implemented, despite the work of the international community to set up rules and practical guidelines (Bonn Guidelines of 2002 and Nagoya protocol of 2010). There are also built-in tensions between the CBD and the framework for trade-related aspects of intellectual property rights in the World Trade Organization texts.

From a social and cultural perspective, the contrasting models mentioned above can be viewed as echoes of the eroding social acceptability of patents on natural resources and related traditional knowledge. In the North, the claims for the right to sow and trade ancient variety of seeds echoes the claims for sovereignty over natural resources in the South. Stakeholders involved in those issues look for alternative models of intellectual property rights.

A new model focusing on partnerships with local communities and on the valuation of traditional knowledge as a non-patentable wealth could be emerging. There is no doubt that such a model is environmentally and socially sustainable. It stills needs to test the conditions for its economic viability, but the examples given in this article are rather encouraging on this aspect. Nevertheless, further research may be necessary to define the economic features of this new business model, which could be to the patent-on-life model what the opensource model is to proprietary software.

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